

JK 12-16 Python First Programmer
Lesson 06 : Loops

Duration : 90 mins

Description

Students learn what is Loop statement and how to use while loop, for loop and infinite loop in Python. They create a few simple projects and two bigger projects by implementing all of these concepts, including “guess the number” game and “math quiz”.

Objectives

Students will be able to:

1. Understand what is loop and use it in a code.
2. Explain the difference between while loop, for loop, and infinite loop and use it in a code.
3. Practice how to design a program flowchart.
4. Analyze code that incorporate loops and the outcome of the code.
5. Create projects combining all concepts learned during the lesson.

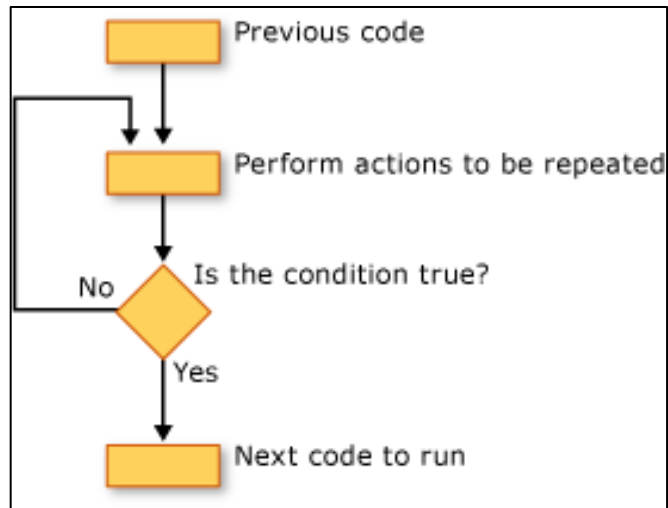
Tools

1. Pycharm

Class Structure

Duration	Topic
5 mins	Preparation
5 mins	Comprehension Check
5 mins	Loops Introduction
15 mins	While Loop
15 mins	For Loop
15 mins	Infinite Loop
10 mins	Flowchart Introduction
15 mins	Math Quiz
5 mins	Wrap Up

Planned Activities	Resources
Preparation (5 mins)	
- Teacher greets students and ask how's their conditions. - Instruct students to open their pycharm. - If possible, teacher can incorporate a mini game during this preparation time to engage with the students.	
Comprehension Check (5 mins)	
- Teacher ask students if they are still remember about the materials from the previous lesson. - Teacher check on students' comprehension regarding the previous lesson by asking relatable questions. - Teacher explain and demonstrate in brief or summarize about what they have learned from the previous lesson.	
Loops Introduction (5 mins)	
Explain - We explain to students, we use loops to repeat the same codes for specified times. - Instead of typing the same code in the same script over and over again, we can use loops to make it more efficient. - The codes inside the loop block will be run until the conditions given is evaluated to be False.	



While Loop (15 mins)

Explain and Show

- While Loop will execute the code inside of it as long as the specified condition is true.
- Inside of while, we need to increment the counter variable of the statement, so it will always be repeated according to the number added.
- The incremented variable inside of the conditional statement will be checked if it's true or not. If it's true, it will do the code inside of the while, but if it's not true, it will not do the code inside of it and move on to the code outside of while.

Explain and Code

- We show the while loop project to students. We explain that without the counter variable and if it's not incremented, the looping will not stop because the condition will always be true.
- We explain and analyze the code of the while loop together with students.

File/Code:

[1.WhileLoop.py](#)

- Tell students to try and change the condition number so we can see the difference.

Explain and Show

- We show the project to students. We explain to students that this program is to receive inputted numbers and multiplication the number for 10 times.
- We tell students to try it by themselves and use the while loop.
- Teachers may give hints to students if they don't understand any of the concepts.

File/Code:

[2.MultiplicativeNumber.py](#)

For Loop (15 mins)

Explain and Show

- With For loop, we need to declare the variable together inside of the loop and specify the range of the counter.
- The first parameter for the range means that we initialize or give value to the counter variable.
- The second parameter for the range means that the counter will be incremented by 1 until less than the second parameter number.

Explain and Code

- We show the For loop project to students. Tell students that the counter will always be incremented by 1 automatically unlike the while loop.
- We explain and code the for loop together with students.
- Tell students to try and change the condition number so we can see the difference.

File/Code:

[3.ForLoop.py](#)

Explain and Show

- We show the project to students. We explain to students that this program is to receive inputted numbers and find the divisors.

File/Code:

[4.DivisorCalculator.py](#)

- Explain to students that divisors are numbers where it can be divided by the inputted number without remains. Teachers may give hints to students by reminding them about modulo.

Infinite Loop (15 mins)

Explain and Show

- Infinite loop is a loop with no end. The program will not stop until a certain condition is fulfilled inside of the loop.
- To create the condition, we can use the conditional statements and add a break to one of the statements.
- **Break** statement is used to terminate the loop and jump out of the closest enclosing loop.

Explain and Code

- We show students how to create an infinite loop with a while loop and put true as the condition.
- Create a conditional statement where in one of the statements, it contains a break statement.

File/Code:

[5.InfiniteLoop.py](#)

Explain and Show

- We show the project to students. We explain to students that this program is guessing the random number.
- If we answer it wrong, the program will always ask us the answer until we are correct. Thus, we need the infinite loop.
- We can give hints to students by saying that this game has the same concept as lesson 4. “How am I?” project. But in this project, we only add the infinite loop.

File/Code:

[6.GuessTheNumber.py](#)

Flowchart Introduction (10 mins)

Explain

- Flowchart is a chart that is used to explain the flow of the program.
- We can give an example by showing the flowchart.jpg. Explain that in flowchart, each shape represents their own function.
- Explain the shapes:
 - **Rounded rectangle**: represent to start or end the program.
 - **Arrow**: the connector to see the direction of the shapes.
 - **Parallelogram**: represent input or output of the program.
 - **Rectangle**: represent a process of the program.
 - **Diamond**: indicates a decision. The decision will result in true or false.

Symbol	Name	Function
	Start/end	An oval represents a start or end point
	Arrows	A line is a connector that shows relationships between the representative shapes
	Input/Output	A parallelogram represents input or output
	Process	A rectangle represents a process
	Decision	A diamond indicates a decision

File/Code:

Flowchart.jpg

Math Quiz (15 mins)

Explain and Show – Flowchart

- We show the output of the project to students.

File/Code:

- Let the students analyze and let them explain the flow of the program.
- We show the flowchart to students and explain the flow of the program.
- The program will random the 2 numbers and an operator. If the operator variable is 1, the question will be added. If the operator variable is 2, the question will be subtraction. If the operator variable is 3, the question will be multiplication.
- The program will not stop unless the user answers the correct answer.

MathQuizFlowChart.jpg

[7.MathQuiz.py](#)

Code

- We tell the students to create the project by themselves based on the flow shown by the flowchart.
- We may give hints to students.

Wrap Up (5 mins)

Wrapping Up

- We sum up by asking questions, making sure the students remember the most important things.
- We thank students for doing a great job overall.

☐ ☆ Good Luck With The Class !! ☆ ☐